

Table 1: Reserve comparison

Economy and reserve	Preparation	Sale	Two admissible bidders	Expected target proceeds	Trading outcome
<i>Panel A. Binary values</i>					
Weak ($r = 1.2$), $p = 0.5$	0.250000	0.687500	0.145833	0.392708	no trade
Weak ($r = 1.2$), $p = 1.01$	0.543573	0.443232	0.053595	0.452756	full orders
Strong ($r = 3$), $p = 0.5$	0.522757	0.920460	0.435631	0.872392	full orders
Strong ($r = 3$), $p = 1.01$	0.513373	0.770226	0.210611	0.987487	full orders
<i>Panel B. Atomless values with class information ($\varepsilon_V = 0.05$)</i>					
Weak ($r = 1.2$), $p = 0.5$	0.250000	0.687500	0.145833	0.392665	no trade
Weak ($r = 1.2$), $p = 1.1$	0.540309	0.391610	0.028025	0.432173	full orders
Strong ($r = 3$), $p = 0.5$	0.522760	0.920460	0.435634	0.872367	full orders
Strong ($r = 3$), $p = 1.1$	0.511638	0.749292	0.200293	1.014500	full orders

Preparation is the probability that the challenger pays its preparation cost. Sale is the probability that at least one bidder submits an admissible bid, whether the incumbent or a prepared challenger whose realized value meets the reserve. Two admissible bidders is the probability that both do. Expected target proceeds are seller revenue per share. All are validated outcome measures of the listed continuation at the declared reserve.

Benchmark primitives, Laplace noise, cost atoms; Panel B draws values uniformly around ℓ and h with half-width $\varepsilon_V = 0.05$ and gives the investor class information only. Analytical support: at every listed node the trading outcome is the unique continuation under the global bound ($k - \Delta_T > 0$ for no trade; the uniform derivative bound with a positive low-cost floor for full orders). Order magnitudes, margins, evidence status, uniqueness scope, and the complete parameter identity of each node are in the online reserve-details table. The exploratory reserve summary (highest revenue among continuations found) is the online exploratory table.

Table 2: Reserve comparisons: orders, margins, evidence, and identity behind Table 4.

Economy and reserve	q_H	q_L	E	A	O_H	$\mathcal{R}_T$	Floor margin	Trading margin	Evidence	Unique
<i>Panel A. Binary values</i>										
Weak ($r = 1.2$), $p = 0.5$	0	0	0.250000	0.250000	0.125000	0.392708	1.73×10^0	3.33×10^{-3}	analytical	yes
Weak ($r = 1.2$), $p = 1.01$	1	-1	0.543573	0.338494	0.338494	0.452756	1.41×10^0	9.08×10^{-3}	analytical	yes
Strong ($r = 3$), $p = 0.5$	1	-1	0.522757	0.522757	0.324192	0.872392	1.37×10^0	2.41×10^{-3}	analytical	yes
Strong ($r = 3$), $p = 1.01$	1	-1	0.513373	0.317504	0.317504	0.987487	1.24×10^0	1.36×10^{-2}	analytical	yes
<i>Panel B. Atomless values with class information</i>										
Weak ($r = 1.2$), $p = 0.5$	0	0	0.250000	0.250000	0.125000	0.392665	1.73×10^0	2.99×10^{-3}	analytical	yes
Weak ($r = 1.2$), $p = 1.1$	1	-1	0.540309	0.336302	0.336302	0.432173	1.39×10^0	1.40×10^{-2}	analytical	yes
Strong ($r = 3$), $p = 0.5$	1	-1	0.522760	0.522760	0.324195	0.872367	1.37×10^0	2.42×10^{-3}	analytical	yes
Strong ($r = 3$), $p = 1.1$	1	-1	0.511638	0.316252	0.316252	1.014500	1.23×10^0	1.38×10^{-2}	analytical	yes

Complete parameter identity of every node: $h=10|ell=1|rho=0.25|c.L=1|c.H=6|b=2|k=0.02|noise=Laplace|cost=atoms|cost_halfwidth=0|prior.H=0.5$ with $r, p, value_law \in \{binary, uniform_classes\}$ and $eps.V \in \{0, 0.05\}$ as listed. Institution, information structure, and tie rule: `reserve_auction:uniform_incumbent[0,r];sale_to_highest_admissible_at_max(p,min(R,v));bid_equal_reserve.is_admissible=yes, public_price_feedback;investor_observes=fundamental_state;buyer_observes=price_only, entry_at_indifference=yes;bid_equal_reserve.is_admissible=yes;upper_plateau_enters_at_tau=M;lower_plateau_enters_at_B(m)=c.L; reserve_auction:uniform_incumbent[0,r];sale_to_highest_admissible_at_max(p,min(R,v));bid_equal_reserve.is_admissible=yes, public_price_feedback;investor_observes=value_class_not_within_class_value;buyer_observes=price_only, entry_at_indifference=yes;bid_equal_reserve.is_admissible=yes;upper_plateau_enters_at_tau=M;lower_plateau_enters_at_B(m)=c.L.`

A is the probability of an admissible prepared challenger. Floor margin is $B_r(m) - c_L$; trading margin is $k - \Delta_T$ for no trade and $(1 - 1/b) e(m) m \Delta_T - k$ for full orders. Evidence is the result status; Unique reports the uniqueness scope recorded with the continuation (unique within all continuations under the stated bound, or not established). Each node is a validated row of `numerics/reserve_events.csv` joined with `tables/reserve_comparisons.csv` on the full parameter identity, the exact reserve, and the branch.

Table 3: Highest revenue among continuations found: exploratory reserve summary with coverage counts.

	Reserve	Node	Found	E range	$\mathcal{R}_T$ range	Search outcome			
<i>Panel A. Highest revenue among continuations found</i>									
Binary values, $r = 1.2$	$p_L (\approx 6.281718)$	event: floor equality p L	1	0.250000	0.785215	analytically unique continuation			
Binary values, $r = 3$	$p_H (\approx 1.325270)$	event: ceiling equal- ity p H	1	0.506477	1.068854	analytically unique continuation			
Class values, $r = 1.2$	$p_L (\approx 6.281718)$	event: floor equality p L	1	0.250000	0.785215	analytically unique continuation			
Class values, $r = 3$	$p_H (\approx 1.325270)$	event: ceiling equal- ity p H	1	0.506477	1.068854	analytically unique continuation			
	Nodes	Events	Several found	Unresolved	None found	Candidates evaluated	Rejected	Cand. unresolved	Duplicates merged
<i>Panel B. Search coverage from the final-pass attempt ledger</i>									
Binary values, $r = 1.2$	392	8	148	116	0	2688	288	116	1744
Binary values, $r = 3$	392	8	148	116	0	2721	321	116	1744
Class values, $r = 1.2$	464	12	155	155	0	3249	382	136	2112
Class values, $r = 3$	416	12	148	116	0	2756	332	116	1744

Highest revenue among continuations found, not an optimal reserve: Panel A selects, for each economy, the node (grid point or exact event) whose highest accepted continuation revenue is largest among the continuations the declared searches found, and reports the ranges of preparation and revenue across the accepted continuations at that node. Range endpoints need not belong to the same continuation; a single value denotes a singleton found range. No global envelope is certified; a node without an accepted continuation is not a nonexistence result, and an unresolved node does not imply an empty equilibrium set. Panel B counts come from the final-pass attempt and validation ledger of `numerics/reserve_ranges.csv`, one record per solved node: nodes attempted, exact-event nodes among them, nodes with several accepted continuations, nodes left unresolved, nodes with no accepted continuation, candidates evaluated, rejected, unresolved, and duplicates merged by the continuation identity. The following exact-event table lists every exact-event node (reserve at zero, at ℓ , at r , at h , at the class-band edges, at the declared reserves, at the published sampled weak reserve, at the weak-incumbent floor equality $p_L = h - c_L/m$, and at the high-cost ceiling equality p_H) with the candidates found there; one-sided offsets from each event are separate nodes counted in Panel B. Earlier attempts that triggered identity refinement are preserved separately in the numerical diagnostics archive. Known limits: the searches cover the declared candidate families and the declared pricing family only (Online Appendix C.6).

Table 4: Exact-event candidates in the exploratory reserve search.

	Event	Reserve	Found	$\mathcal{R}_T$ max	Unresolved	Search outcome
4	Binary, $r = 1.2$	reserve zero	0	1 0.147917	no	analytically unique continuation
	Binary, $r = 1.2$	declared reserve 0.5	0.5	1 0.392708	no	analytically unique continuation
	Binary, $r = 1.2$	reserve equals ell	1	1 0.377083	no	analytically unique continuation
	Binary, $r = 1.2$	declared reserve 1.01	1.01	1 0.452756	no	analytically unique continuation
	Binary, $r = 1.2$	reserve equals r	1.2	1 0.400229	no	analytically unique continuation
	Binary, $r = 1.2$	ceiling equality p H	$p_H (\approx 1.691903)$	1 0.535726	no	analytically unique continuation
	Binary, $r = 1.2$	floor equality p L	$p_L (\approx 6.281718)$	1 0.785215	no	analytically unique continuation
	Binary, $r = 1.2$	reserve equals h	10	1 0.000000	no	accepted continuation(s) found; uniqueness not established
	Binary, $r = 3$	reserve zero	0	1 0.655633	no	analytically unique continuation
	Binary, $r = 3$	declared reserve 0.5	0.5	1 0.872392	no	analytically unique continuation
	Binary, $r = 3$	reserve equals ell	1	1 1.049608	no	analytically unique continuation
	Binary, $r = 3$	declared reserve 1.01	1.01	1 0.987487	no	analytically unique continuation
	Binary, $r = 3$	ceiling equality p H	$p_H (\approx 1.325270)$	1 1.068854	no	analytically unique continuation
	Binary, $r = 3$	reserve equals r	3	1 0.375000	no	analytically unique continuation
	Binary, $r = 3$	floor equality p L	$p_L (\approx 6.281718)$	1 0.785215	no	analytically unique continuation
	Binary, $r = 3$	reserve equals h	10	1 0.000000	no	accepted continuation(s) found; uniqueness not established
	Classes, $r = 1.2$	reserve zero	0	1 0.147873	no	analytically unique continuation
	Classes, $r = 1.2$	declared reserve 0.5	0.5	1 0.392665	no	analytically unique continuation
	Classes, $r = 1.2$	reserve equals ell minus epsV	0.95	1 0.390321	no	analytically unique continuation
	Classes, $r = 1.2$	reserve equals ell	1	1 0.540618	yes	unresolved search (open nodes or unconverged starts retained)
	Classes, $r = 1.2$	reserve equals ell plus epsV	1.05	1 0.444542	no	analytically unique continuation
	Classes, $r = 1.2$	declared reserve 1.1	1.1	1 0.432173	no	analytically unique continuation
	Classes, $r = 1.2$	reserve equals r	1.2	1 0.400229	no	analytically unique continuation
	Classes, $r = 1.2$	ceiling equality p H	$p_H (\approx 1.691903)$	1 0.535726	no	analytically unique continuation
	Classes, $r = 1.2$	floor equality p L	$p_L (\approx 6.281718)$	1 0.785215	no	analytically unique continuation

Continued on next page

Table 4: Exact-event candidates in the exploratory reserve search (continued).

	Event	Reserve	Found	$\mathcal{R}_T$ max	Unresolved	Search outcome
Classes, $r = 1.2$	reserve equals h minus epsV	9.95	1	0.000000	no	accepted continuation(s) found; uniqueness not established
Classes, $r = 1.2$	reserve equals h	10	1	0.000000	no	accepted continuation(s) found; uniqueness not established
Classes, $r = 1.2$	reserve equals h plus epsV	10.05	1	0.000000	no	analytically unique continuation
Classes, $r = 3$	reserve zero	0	1	0.655609	no	analytically unique continuation
Classes, $r = 3$	declared reserve 0.5	0.5	1	0.872367	no	analytically unique continuation
Classes, $r = 3$	reserve equals ell minus epsV	0.95	1	1.033719	no	analytically unique continuation
Classes, $r = 3$	reserve equals ell	1	1	1.018643	no	analytically unique continuation
Classes, $r = 3$	reserve equals ell plus epsV	1.05	1	0.999860	no	analytically unique continuation
Classes, $r = 3$	declared reserve 1.1	1.1	1	1.014500	no	analytically unique continuation
Classes, $r = 3$	ceiling equality p H	$p_H (\approx 1.325270)$	1	1.068854	no	analytically unique continuation
Classes, $r = 3$	reserve equals r	3	1	0.375000	no	analytically unique continuation
Classes, $r = 3$	floor equality p L	$p_L (\approx 6.281718)$	1	0.785215	no	analytically unique continuation
Classes, $r = 3$	reserve equals h minus epsV	9.95	1	0.000000	no	accepted continuation(s) found; uniqueness not established
Classes, $r = 3$	reserve equals h	10	1	0.000000	no	accepted continuation(s) found; uniqueness not established
Classes, $r = 3$	reserve equals h plus epsV	10.05	1	0.000000	no	analytically unique continuation